

Zelong Xu

EDUCATIONAL BACKGROUND

University of Wisconsin-Madison

B.S. in Statistics (expected Dec 2026)

Certificates: Computer Science, Mathematics

- GPA: 3.90/4.00 (*Dean's List, 3 semesters*)
- Core Courses: Interpretable Machine Learning, Matrix Methods in Machine Learning, Deep Learning, Numerical Analysis

Shandong University (*transferred to UW-Madison*)

Sep 2024 – Dec 2026

Sep 2022 – Jun 2024

- GPA: 88.1/100

PUBLICATIONS

DOG-DPO: Dynamic Optimization in Geometry for Safety Alignment

Yi Nian, Tiankai Yang, Yudi Zhang, Qi Pan, **Zelong Xu**, Shenzhe Zhu, Qingqing Luan, Yue Huang, Xiangliang Zhang, Yue Zhao.

EMNLP 2026 · Advisor: Prof. Yue Zhao · [arXiv]

- Co-developed DOG-DPO's geometric approach to preference-data selection, exploring alternative designs and conducting core experiments to assess the resulting safety-utility trade-off.
- Using 11.6% of preference pairs, DOG-DPO recovered most full-data DPO safety gains across two backbones and six safety and utility benchmarks, with 15–35× faster data selection than representative baselines.

PREPRINTS

Statistical Confidence for LLM Circuit Discovery via Randomized Interventions

Zelong Xu, Yang Lu.

Under review, 2026 · Advisor: Prof. Yang Lu

- Developed a randomized-intervention framework that estimates and tests edge contributions across circuit densities, sharing model evaluations across candidate edges to make statistical inference practical at scale.
- Validated the framework across five model-task settings with up to 34,234 candidate edges; p-values were uniform or conservative on structural negative controls, with zero discoveries after multiple-testing correction.

CUA-SWE: When Computer-Use Agents Meet Visual Software Engineering

Prince Zizhuang Wang*, Chenhao Liang*, **Zelong Xu***, Aojie Yuan*, Xiaolin Zhou*, Haiyue Zhang, Yue Zhao, Xiyang Hu, Shuli Jiang.

*Equal contribution.

Under review, 2026 · Advisors: Prof. Yue Zhao and Prof. Xiyang Hu · [arXiv]

- Designed and built all 20 DevOps tasks for CUA-SWE, a 105-task benchmark for visual software engineering; developed automated task verifiers to assess whether agents produce correct software changes in running environments.
- Produced the paper's figures, project website, and demos to explain the benchmark design and illustrate how agents inspect software, diagnose failures, and verify repairs.

SyncRA: Learning Temporal Correspondence in Omni-Modal Models

Zelong Xu, Yan Li, Wenhe Hu, Xiyang Hu.

Under review, 2026 · Advisor: Prof. Xiyang Hu · [arXiv]

- Diagnosed failures in audio-visual temporal correspondence through controlled temporal swaps; proposed SyncRA to align intermediate audio and video representations through within-video contrastive learning supervised by input timing.
- Across four omni-modal backbones and five video benchmarks, SyncRA outperformed answer-only fine-tuning in all 20 model-benchmark combinations and improved temporal-swap accuracy, with unchanged inference.

OTHER RESEARCH EXPERIENCE

EmoPair: Counterfactual Paired Listwise Alignment for Emotion-Aware Speech LLMs *Nov 2025 – May 2026*

- Developed EmoPair to align speech-model responses with vocal emotion, combining same-text speech variants, listwise preference learning, and paired comparisons against neutral speech.
- Outperformed GRPO baselines on IEMOCAP and held-out RAVDESS, improving response appropriateness from 72.4 to 77.2 and 47.9 to 55.8, respectively.

BR-LOCAL: A Solver-Verified Benchmark for Ripple Effects under In-Context Knowledge Editing *Oct 2025 – Apr 2026*

- Designed BR-LOCAL to measure how in-context knowledge edits propagate through dependent facts, using synthetic Horn knowledge bases and solver-verified gold change-sets.
- Built a procedural generator and evaluation harness for 5K episodes, with paired cold-start and online tracks to compare static reasoning with in-context edit propagation.

Multi-Emotional Probing for Controllable Language Generation *Mar 2025 – May 2025*

- Developed a probe-and-inject framework for emotion-conditioned language generation, extracting emotion directions from BERT representations and using vector injection to steer Llama-2 outputs.
- Analyzed layerwise emotion representations, identifying concentration in middle layers and near-orthogonal directions; evaluated the framework on targeted emotions including joy and anger.

INTERNSHIP EXPERIENCE

Icarus Fund, Remote *Jul 2024 – Aug 2024*
Quantitative Analyst

- Developed and backtested Python-based quantitative models for S&P 500 companies, optimizing portfolio strategies with Sharpe ratios up to 1.42.

SKILLS & LANGUAGES

Technical Skills: Python, Java, JavaScript, R, SQL; PyTorch, Hugging Face Transformers, TransformerLens, scikit-learn, vLLM; multi-GPU LLM training

Languages: Chinese (Native), English (Fluent, TOEFL: 109/120)